MIL-E-13191B
26 February 1969
SUPERSEDING
MIL-E-13191A
26 January 1956

MILITARY SPECIFICATION

ELECTRODES, WELDING, COVERED, BRONZE: FOR GENERAL USE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers bronze covered electrodes for general welding of high-strength bronzes and brasses to each other, or to dissimilar metals (see 6.1).

1.2 Classification Electrodes covered by this specification shall be of the following classes and sizes, table I, as specified in the contract or order (see 6.2).

Table I. Classes and sizes of electrodes

Class	Size, (Diameter of core wire) inch	Length, inches
MIL-ECuSi (silicon-bronze)	3/32	9 and 11
	1/8, 5/32, 3/16, 1/4	14
	1/4	18
MIL-ECuSn-A (phosphor-bronze)	3/32	9
	1/8, 5/32, 3/16, 1/4	14
	1/8, 5/32, 3/16, 1/4	18
MIL-ECuSn-C (phosphor-bronze)	3/32	9
	1/8, 5/32, 3/16, 1/4	14

FSC 3439

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2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-B-750 - Bronze, Phosphor, Bar, Plate, Rod, Sheet, Strip, Flat Wire, and Structural and Special Shaped Sections

QQ-C-591 - Copper-Silicon Alloy, Rod, Wire, Shapes, and Flat Products (Flat Wire, Strip, Sheet, Bar, and Plate)

MILITARY

MIL-W-10430 - Welding Rods and Electrodes, Preparation for Delivery of

STANDARDS

FEDERAL

Fed. Test Method Std. No. 151 - Metals, Test Methods

MILITARY

MIL-STD-20 - Welding Terms and Definitions
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN WELDING SOCIETY

AWS A2.0-68 - Welding Symbols

(Application for copies should be addressed to the American Welding Society, 345 E. 47th Street, New York, N. Y. 10017).

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3. REQUIREMENTS

3.1 Chemical composition.

3.1.1 Core wire. The chemical composition of the core wire shall conform to the requirements specified for their classification in table II, as determined in accordance with 4.4.1.

3.2 Coverings of electrodes.

3.2.1 Handling The coverings shall be capable of withstanding ordinary handling without being injured.

3.2.2 Concentricity of coverings The coverings shall be concentric to the extent that for all sizes of electrodes the maximum core-plus-one covering dimension shall not exceed the minimum core-plus-one covering dimension by more than 5 percent of the mean dimension (see 4.4.2).

3.2.3 Uniformity of consumption. Coverings shall be such as to be consumed uniformly (see 4.4.2).

3.2.4 Flaking. The coverings shall not exhibit flaking at any time while electrodes are being used with the respective maximum currents for general welding recommended by the manufacturer, nor shall the coverings exhibit flaking upon cooling after approximately half of the electrode has been so consumed or upon resumption of welding.

3.2.5 Stability in storage. The coverings shall be such that electrodes shall be capable of complying with this specification after storage under roof and on dry floors or platforms for periods up to 12 months in original unopened containers.

3.2.6 Toxic fumes. The fumes from the burning coverings shall not be injurious to personnel when electrodes are used in adequately ventilated spaces.

3.2.7 Slag removal The slag produced by the coverings during welding shall be readily removable with hand tools (not air-or power-operated) from weld deposits that have cooled nearly to body temperature or approximately 100°F.

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Table II. Chemical composition of core wire - percent[1]

Class	Copper	Aluminum	Iron	Manganese	Phosphorus	Lead	Silicon	Tin	Zinc	Total other elements[3]
MIL-ECuSi	Rem.	0.01*	0.5	1.5[2]	(*)	0.02*	2.8-4.0	1.5[2]	(*)	0.50
MIL-ECuSn-A	Rem.	0.01*	(*)	(*)	0.15-0.35	0.02*	(*)	4.8-5.8	(*)	0.50
MIL-ECuSn-C	Rem.	0.01*	(*)	(*)	0.35	0.02*	(*)	7.0-9.0	(*)	0.50

[1]Single values shown are maximum
[2]One or both of these elements may be present within the limits specified.
[3]Total other elements, including the elements marked with an asterisk (*) shall not exceed the value specified.

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3.2.8 Extent of coverings.

3.2.8.1 Arc ends. The arc ends of the electrodes shall be sufficiently bare to permit easy striking of the arc. The length of the bare portion of the arc end (measured from the end of the point where the full cross section of the covering is obtained) shall not exceed the diameter of the core wire or one-eighth inch, whichever is less. (Electrodes with chipped coverings near the arc end, baring the core wire slightly more than the specified distance, may be acceptable provided no chip uncovers more than 50 percent of the circumference of the core.)

3.2.8.2 Grip ends. Not less than 1/2 inch of the grip ends of the electrodes shall be bare to permit electrodes to be readily inserted in holders, but not more than 1 1/4 inches shall be bare in order that the stub waste need not exceed 1 3/4 inches.

3.3 Dimensions.

3.3.1 Diameter. The diameter of the wire of the electrode shall not vary from the specified diameter more than plus or minus 0.002 inch.

3.3.2 Lengths. The nominal length of the core wire shall be as specified in table I for the sizes as listed (see 1.2). The length of any electrode shall not vary more than 3/16 inch from the approved nominal length.

3.4 Mechanical properties.

3.4.1 Tensile strength. The tensile strength shall meet the requirements of table III when prepared as specified in 4.4.2.1.3.

Table III. Tensile strength requirements (minimum, psi)

Class	Transverse
MIL-ECuSi	50,000
MIL-ECuSn-A	35,000
MIL-ECuSn-C	40,000

3.4.2 Bend ductility. A guided-bend specimen (both root and face) shall be tested in accordance with 4.4.3.2 and shall show no cracks or open defects exceeding 1/8 inch measured in any direction.

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3.5 Character of weld metal. Under the conditions specified in 4.4.4, electrodes shall deposit weld metal that will not exhibit undercutting or overlapping (rolling) and that will be free of slag entrapment, weld-metal cracking, and porosity.

3.6 Electrode marking. Grip-ends of electrodes shall be imprinted on the grip length of each electrode. The printed designation shall be readable under normal lighting conditions.

3.7 Workmanship.

3.7.1 Core wire. The core wires shall be free of injurious defects, such as segregations and nonmetallic inclusions, and may be made by any process yielding a product suitable for the purpose intended, and conforming to the requirements of this specification.

3.7.2 Covering. The coverings of the electrodes shall be free of injurious scabs, blisters, pockmarks, bruises, and any other defects that may affect the serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. A lot shall consist of 5,000 pounds or fraction thereof of the same size and class.

4.3 Sampling.

4.3.1 For chemical analysis, coverings of electrodes, dimensions, mechanical properties, toxic fumes, character of weld-metal and visual identification. A random sample of 10 pounds shall be selected for the above tests. Whenever possible, the same electrode may be used for several tests.

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4.4 Test procedures.

4.4.1 Chemical analysis of core wire. Chemical analysis shall be conducted on a minimum sample of 2 ounces in accordance with method 111 or 112 of Fed. Test Method Std. No. 151.

4.4.2 Covering of electrode.

4.4.2.1 Concentricity of covering. The concentricity and average thickness of the coverings on the three electrodes of each size shall be measured by either of the methods described in 4.4.2.1.1 or 4.4.2.1.2.

4.4.2.1.1 Micrometer method. A small amount of covering shall be entirely removed from a spot on one side of the core wire near one end of the covered length, care being taken to insure that no metal is removed from the core wire. The diameter of the core wire plus the thickness of the covering on the side opposite the bared spot shall be measured with a micrometer. The covering shall then be removed from a spot on the opposite side of the core wire at a point immediately adjacent to that at which the first measurement was made, and a second similar measurement made. Second and third pairs of similar measurements shall be made at approximately the mid-length and the other end of the covered portion, on diameters approximately 60° and 120°, respectively, from the diameter on which the first pair of measurements was made. The pair of measurements (two adjacent measurements) that show the greatest variation shall be used to determine the acceptability of the electrodes as specified in 3.2.2.

4.4.2.1.2 Micrometer method using supplemental strip (alternate method). At a point near one end of the covered length of the electrode, a portion of the covering material in the form of a band approximately 1/2 inch wide shall be removed, care being taken to insure that no metal is removed from the core wire. Using a micrometer and a supplemental metal strip (approx. 1 inch long by 1/2 inch wide by any convenient and accurately measured thickness, e.g., 0.125 or 0.250 inch or a specially adapted micrometer having an anvil with such a strip permanently attached thereto), the diameter of the core wire plus the thickness of the covering on one side of the electrode shall be measured. The metal strip shall bridge the gap left by the removal of the band of covering material, and the core-wire diameter plus the thickness of the covering on one side will be equal to the total measurement minus the thickness of the metal strip. Several measurements shall be made on several different diameters around the area from which the covering was removed, and the maximum and minimum measure-

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ments shall be noted. A second and third series of similar measurements shall be made at approximately mid-length and other end of the covered portion, respectively. The maximum and minimum dimensions at any one section that show the greatest difference shall be used to determine the acceptability of the electrodes as specified in 3.2.2.

4.4.2.1.3 Operating characteristics. The operating characteristics of electrodes shall be tested by welding a test assembly for transverse-tension tests (4.4.3.1) and guided-bend tests (4.4.3.2) using welding techniques specified by the manufacturer of the particular electrode nder test. Standards AWS-A2.0-6.8 and MIL-STD-20 shall apply. The base metal to be used for test plates shall conform to the specifications prescribed in table IV for the class of electrode to be tested. All welding shall be done in the flat position. The test joints shall conform to the dimensional requirements shown in figure 1. The test assembly shall be restrained sufficiently during welding to prevent warping in excess of 5°. A test assembly that has warped more than 5° shall be discarded. Test plates shall not be straightened. Test assemblies shall be within the temperature ranges shown in table V before starting to weld each pass. If, after any pass, the maximum temperature specified is exceeded, plates shall be allowed to cool in air to within the specified temperature range. Water shall not be used for cooling.

Table IV. Base metal for test specimens: Transverse tension (all-weld metal) and guided bend

Class	Base metal
MIL-ECuSi	QQ-C-591, Copper Alloy No. 655, Annealed
MIL-ECuSn-A	QQ-B-750, Composition A, soft (Copper Alloy No. 510)
MIL-ECuSn-C	QQ-B-750, Composition A, soft (Copper Alloy No. 510)

Table V. Preheat and interpass-temperature ranges

Class	Min. °F	Max., °F
MIL-ECuSi	60	150
MIL-ECuSn-A	350	550
MIL-ECuSn-C	350	550

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4.4.3 Mechanical properties.

4.4.3.1 Transverse-tension test. Two transverse-tension test specimens shall be machined from the test assembly (4.4.2.1.3), shall conform to the dimensional requirements shown in figure 2, and shall be subjected to tension load until failure. The mechanical properties of each of the two tension test specimens shall conform to the requirements prescribed in table III.

4.4.3.2 Guided-bend test. One face-bend and one-root-bend test specimen shall be machined from the test assembly (4.4.2.1.3) and shall conform to the dimensional requirements shown in figure 3. Each specimen shall be bent in a jig having the working contour shown in figure 4 and otherwise made substantially in accordance with figure 4. Any convenient means may be used for moving the plunger member in relation to the die member. The specimen shall be placed on the die member with the weld at mid-span. Face-bend specimens shall be placed with the face of the weld toward the gap. Root-bend specimens shall be placed with the root of the weld towards the gap. The two jig members shall be forced together until the specimen conforms to a U-shape and until a 1/32 inch diameter wire can not be placed between the specimen and any point on the curvature of the plunger member. The specimen shall then be taken out of the jig and the convex surface examined. Both specimens shall show no cracks or other open defect exceeding 1/8 inch measured in any direction. Cracks occuring on corners of the specimen shall not be considered.

4.4.4 Visual examination. The weld deposit of the test assembly (4.4.2.1.3) shall be examined for surface gas pockets, smoothness of surface, undercutting, overlapping, and cracking.

4.4.5 Toxic performance. The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the procuring activity to the appropriate department medical service who will act as an advisor to the procuring agency.

4.4.6 Preparation for delivery. Prior to shipment, examination shall be made to determine compliance with section 5.

5. PREPARATION FOR DELIVERY

5.1 Levels of packaging. The level of packaging shall be level A or C unless otherwise specified (see 6.2).

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5.1.1 Level A. Unless otherwise specified in the contract or purchase order, electrodes shall be packaged in accordance with MIL-W-10430. A unit package shall contain 5, 10, 25, or 50 pounds of electrodes of one size and type as specified by the procuring activity.

5.1.2 Level C. Unless otherwise specified in the contract or purchase order, electrodes shall be packaged in class 1 or class 2 containers in accordance with MIL-W-10430.

5.2 Packing.

5.2.1 Electrodes, packaged in accordance with level A or C as specified, shall be packed in accordance with level A, B, or C of MIL-W-10430, as specified, (see 6.2).

5.3 Tests on shipping containers as specified in MIL-W-10430 shall apply to level A and B packing only.

5.4 Marking of shipment. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Electrodes herein specified are for joining copper alloys and steel, as well as for building up surfaces. Electrodes are designed for use with direct current, reverse polarity (electrode positive).

6.1.1 Class MIL-ECu-Si. These electrodes are used primarily for welding copper, copper-silicon, and copper-zinc (brass) base metals to themselves. The use of string beads or narrow weld passes reduces contraction stresses and also permits faster cooling through the hot-short range. Silicon-bronze deposits are seldom used for overlaying bearing surfaces but are often used on areas subject to corrosion.

6.1.2 Class MIL-ECuSn-A and MIL-ECuSn-C. These electrodes are used for joining phosphor bronzes of similar composition and may be employed interchangeably for joining copper, brasses, bronzes, cast iron and other ferrous metals, and dissimilar metals. Phosphor-bronze electrodes have a tendency to flow sluggishly, requiring preheat and interpass temperatures of at least 350°F, especially on heavy sections. Post-weld heat treatment may not be necessary but is desirable for maximum ductility, particularly if the weld metal is cold-worked. Moisture, both on the work and in the electrode coverings, must be strictly avoided. Baking the covered electrodes at 250° to 300°F before use may be necessary.

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6.1.2.1 Electrodes conforming to classification MIL-ECuSn-A are used primarily for joining plates of similar composition in the fabrication of pressure vessels and chemical processing equipment. They may also be used for welding copper if the resultant weld metal is satisfactory for the application as to electrical conductivity and corrosion resistance. Other applications include the joining of copper and copper alloys to mild steel and the braze-welding of hardenable low-alloy steels.

6.1.2.2 The electrodes of higher tin content (classification MIL-ECuSn-C) are particularly useful in providing weld metal of greater hardness than obtained with MIL-ECuSn-A electrodes. The MIL-ECuSn-C electrodes also provide higher tensile and yield strengths than those of the lower tin content of MIL-ECuSn-A electrodes.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class and size (see table I) of electrodes required.
(c) Whether packaging should be level A or C (see 5.1.1 or 5.1.2) and whether packing should be by level A, B or C (see 5.2).

6.2.1 Electrodes should be ordered by the pound.

6.3 Transfer. In determining compliance with these requirements, the tendency of molten metal from bronze electrodes to "sluff" off rather than transfer smoothly, as does that from steel electrodes, should not be interpreted as a sputtering or irregular arc.

6.4 Capable of. The term "capable of" as used in this specification (see 3.2.5) means that the test need not be performed by the producer, however, should subsequent testing by the procuring activity establish that the material does not meet these requirements, the lot will be rejected.

Custodian:
Army - MU
Navy - YD
Air Force - 84

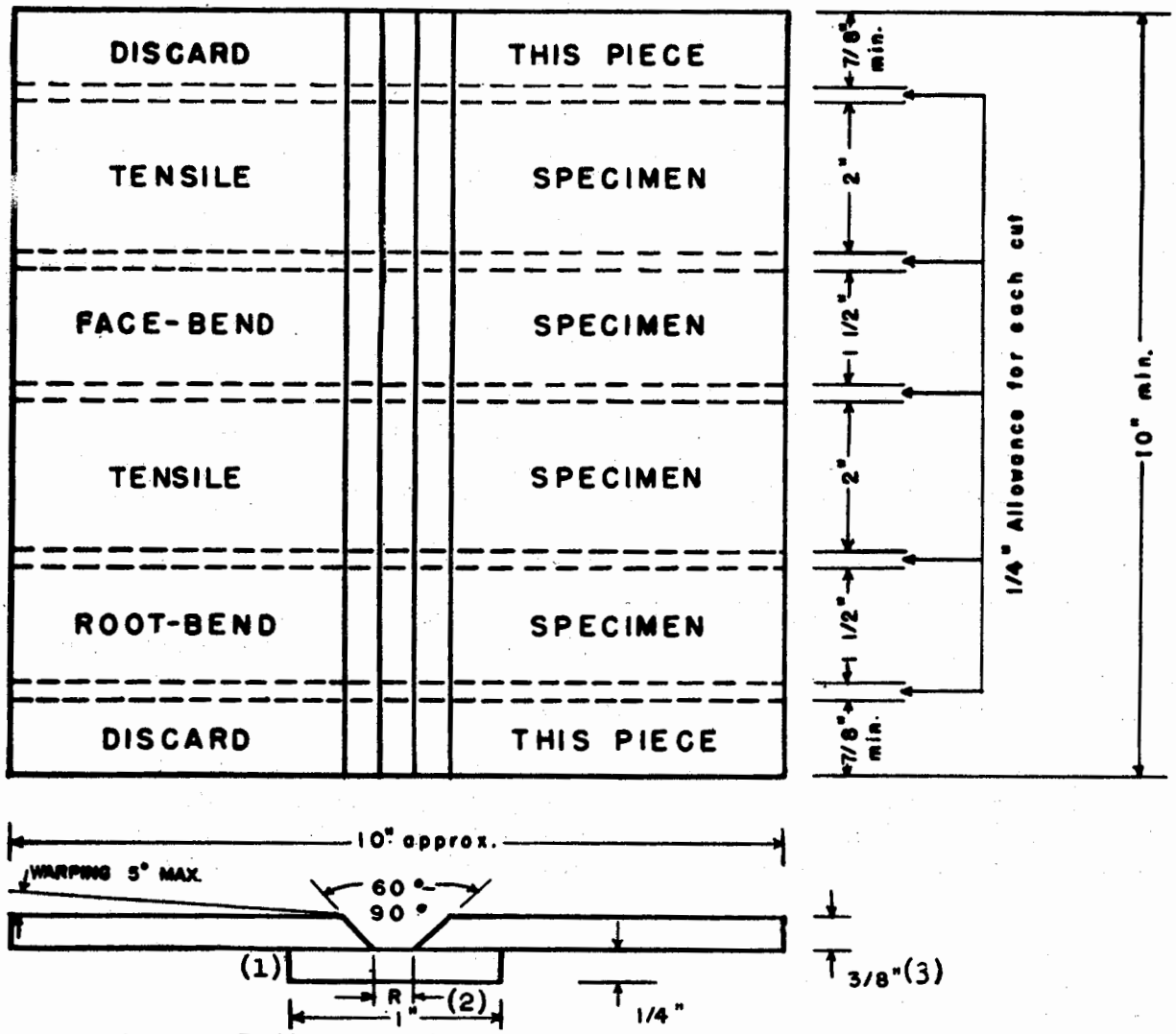
Preparing activity:
Army - MU

Project No. 3439-0130

Review activities:
Army - MU, MO, WC
Air Force - 84, 85
Navy - YD

User activities:
Navy - SH, OS
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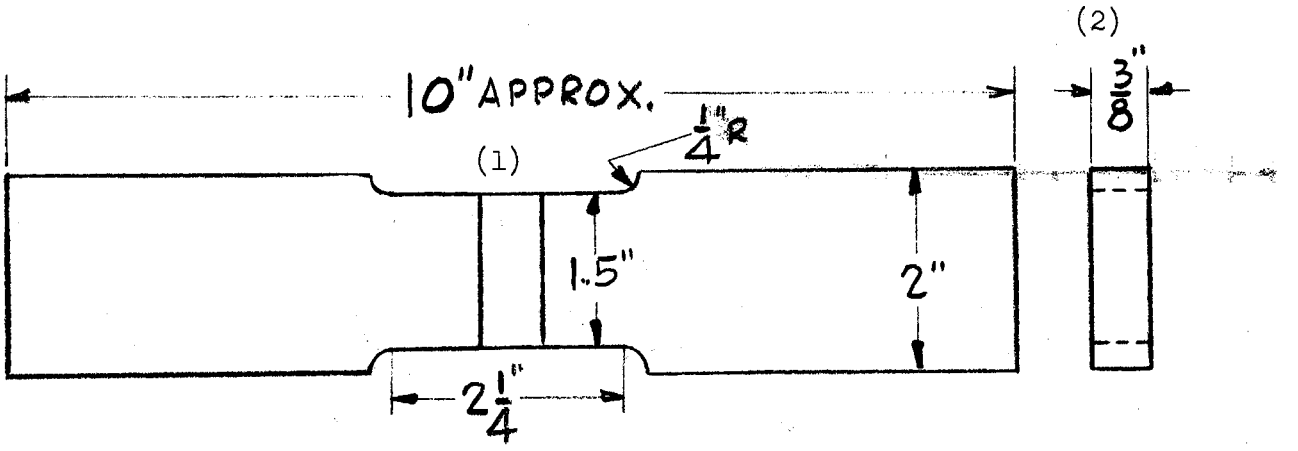
Notes. 1. Backing may be of same material as base metal; copper or carbon may also be used.

2. Root opening shall be same size as electrode.

3. For electrodes ⅛ and less, 3/16-inch plate shall be used.

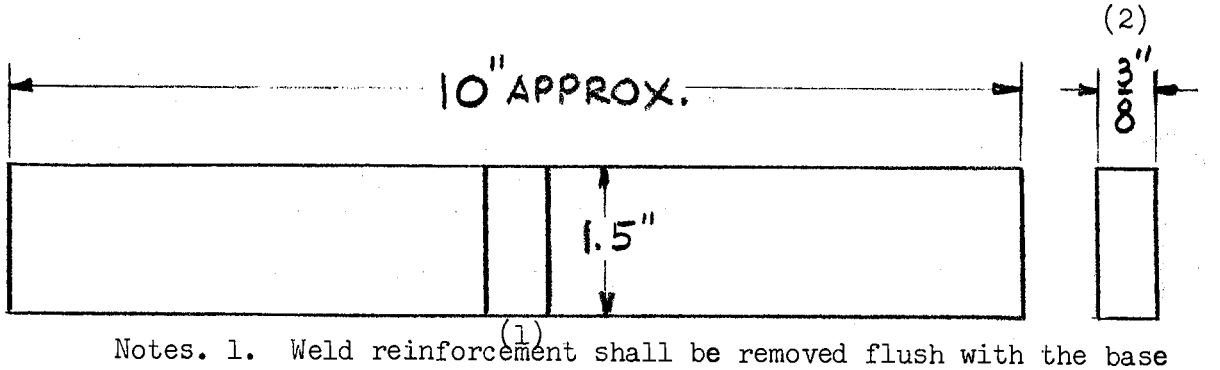
FIGURE 1. *Test assembly.*

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Notes. 1. Weld reinforcement shall be removed flush with the base metal; undercut shall not be removed. Final machining shall be in a direction transverse to the weld
2. For electrodes 1/8 inch and less, 3/16-inch plate shall be used.

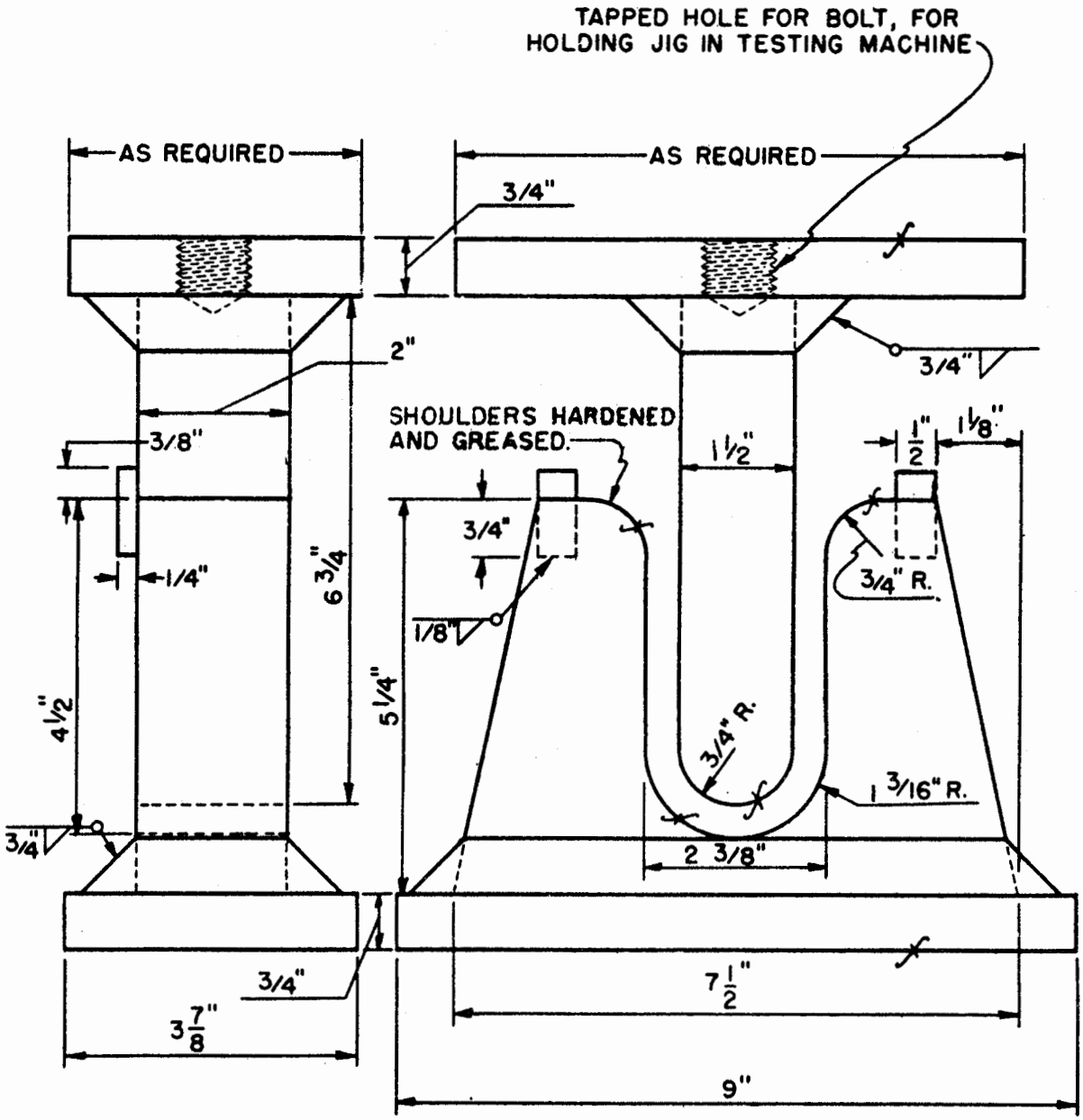
Figure 2. Transverse tension test specimens



Notes. 1. Weld reinforcement shall be removed flush with the base metal; undercut shall not be removed. Final machining shall be done transverse to the weld.
2. For electrodes less than 1/8 inch, 3/16-inch plate shall be used.

Figure 3. Guided-bend test specimen.

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Notes. 1. All dimensions other than those of the working contour, are for information only.

2. Hardened rolls may be used on shoulders, if desired.

3. Specific dimensions are for ⅜-inch specimen, (also for 3/16).

FIGURE 4. *Jig for guided-bend test.*

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